COMPARATIVE STUDY OF 4 GNN ARCHITECTURES FOR B-CELL CONFORMATIONAL EPITOPE PREDICTION

Alexandre Guenegan
Efrei Paris
Panthéon-Assas University

Simón Mariño Perea
BISITE
University of Salamanca

ABSTRACT

Predicting accurately the position of conformational epitope in B-cell proteins would have great consequences on vaccine design and therapeutics. The recent and promising approaches use graph based deep learning models, generally the Graph Neural Networks (GNN) framework. In this study we compare several architectures such as Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), Multi-view Graph Neural Networks (MGAT) and Equivariant Graph Neural Networks (EGNN). We show that using a K-nearest neighbor graph can benefit to the performance for non-attentive models, that neighbor selection is key, and that the use of raw euclidean distance on edges does not improve predictions.

Keywords: Conformational, B-cell epitope prediction, Graph Neural Networks, ESM

1 Introduction

B-cell proteins play a fundamental role in the body's defense against pathogens. They can generate an appropriate immunologic response when triggered by a specific antigen (Ag), and this detection occurs when the antibody (Ab) binds to specific regions of Ag. These regions are known as B-cell epitopes and act as key to the development of vaccines [1]. Epitopes can be classified as linear or conformational, linear epitopes are continuous in the amino acid sequence, while conformational epitopes are discontinuous but spatially closely located. Although the majority of epitopes are conformational, most of the Ab-agnostic methods mainly focus on linear epitopes, and the few tools that attempt to predict discontinuous epitopes show poor performance [2, 3].

Identifying conformational epitopes is a complex challenge because it requires the use of models that are closest to the representation of proteins. Recent works use the GNN framework to address this problem, as proteins can be represented with a graph, this approach showed great results [4, 5]. In this context, we present a comparison of 4 different GNN models to evaluate the impact of topological properties and characteristics of features. Our contributions are as follows:

1. A comparison of four GNN families (GCN, GATv2, MGAT, EGNN) on the Epitope3D benchmark under identical training conditions.
2. An ablation study on graph topology (distance-threshold *vs.* k-nearest neighbors).
3. An ablation study of edge features (with and without Euclidean distances).
4. The introduction of an attention-based EGNN variant that selectively weights messages while preserving E(3)-equivariance.
5. A multi-view aggregation strategy combining four heterogeneous graph topologies via MGAT.

2 Materials and Methods

2.1 Dataset

We use the **Epitope3D** benchmark dataset [2], published in 2022 and originally constructed for the Epitope3D model. Thanks to its rigorous curation, it has since become a standard benchmark for conformational epitope prediction. Antigens were extracted from antibody–antigen complexes deposited in the Protein Data Bank (PDB) before May 2021, under the assumption that epitope regions from multiple complexes can be aggregated onto the same antigen chain. A residue is labelled as *epitope* if it lies within 4 Å of any antibody residue, a definition frequently adopted in the literature. The cleaning phase has 3 steps: (i) a sequence-similarity threshold retaining only antigens with less than 70% identity, (ii) structural consistency verification with PyMOL, and (iii) removal of residues with a relative solvent accessibility (RSA) of zero, since fully buried residues cannot participate in antibody binding. Ultimately the dataset contains 245 antigen structures comprising 168 739 amino acids, of which only 3.56% are epitope residues, split into training (180), validation (20), and blind test (45) sets.

Node features Each residue i in a protein is represented by a characteristic vector $\mathbf{h}_i \in \mathbb{R}^{1792}$, obtained by concatenating two pre-trained protein language model (PLM) embeddings: ESM-2 (650M) [6] and ESM-IF1 [7]. The literature has already shown empirically that the combination of several residue descriptors helps the model in its prediction because of the extended context it gives, especially with the use of both structural and evolutionary PLM [2, 5].

2.2 Graph construction

A protein is represented as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $v_i \in \mathcal{V}$ corresponds to a residue and edges $e_{ij} \in \mathcal{E}$ encode spatial or sequential proximity. The 3-D position of each node is defined by its C_α atom coordinates $\mathbf{x}_i \in \mathbb{R}^3$.

Based on the approach used for Epiformer [4], we generated 4 graphs for each protein to depict different aspect of it:

- **Spatial$_1$ (distance threshold):** An edge is created between every pair of residue whose distance is at most 10 Å.
- **Spatial$_2$ (distance threshold):** Each residue is connected to its K=15 closest neighbors. Epigraph [5] showed experimentally that homophily is an inner property of residues in conformational epitope, using a graph based on clusters might help the model to use this property,
- **Sequential$_1$ and Sequential$_2$ (sequence neighbours):** Edges connect residues that are one or two positions apart in the primary sequence, respectively.

2.3 Training protocol

Loss functions Two loss functions have been tested:

1. **Binary Cross-Entropy (BCE)** with logits, applied after masking buried residues.
2. **Focal Loss** [8], parameterised with $\alpha = 0.25$ and $\gamma = 2$:

$$\mathcal{L}_{\text{focal}} = -\alpha_t \, (1 - p_t)^\gamma \, \log(p_t), \tag{1}$$

where p_t is the model's estimated probability for the correct class and α_t takes the value α for positive (epitope) samples and $1 - \alpha$ for negatives. By down-weighting well-classified (easy) examples, focal loss concentrates the gradient on the rare, hard-to-classify epitope residues.

Cross validation The training and validation sets are merged into a pool of 200 proteins, which is divided into 10 stratified folds. In each fold, 80% of the proteins are used for training and 20% for validation. An early stopping criterion laying on the augmentation of AUC-PR is used, and the model kept is the last one before a stagnation or a loss of AUC-PR is detected on the validation set. Then the threshold is maximized using the Matthews Correlation Coefficient (MCC) on the validation set. Same hyperparameters were used for every model.

3 Model Architectures

We now detail each architecture, highlighting the design choices motivated by the epitope prediction setting.

3.1 Graph Convolutional Network

Background. The Graph Convolutional Network (GCN) [9] is especially powerful because of its simplicity. At layer ℓ, the node update rule is:

$$\mathbf{H}^{(\ell+1)} = \sigma\big(\tilde{\mathbf{D}}^{-\frac{1}{2}}\,\tilde{\mathbf{A}}\,\tilde{\mathbf{D}}^{-\frac{1}{2}}\,\mathbf{H}^{(\ell)}\,\mathbf{W}^{(\ell)}\big), \tag{2}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with added self-loops, $\tilde{\mathbf{D}}$ is the corresponding degree matrix, $\mathbf{W}^{(\ell)}$ is a learnable weight matrix, and σ is a non-linear activation (ReLU).

This model is especially interesting for our problem as it uses a similar coefficient of 1 on each neighbor during message passing. However, we can modify this by applying a single coefficient, such as distance, in order to primarily compare the impact of a coefficient on edges.

Edge distance ablation. When the `GCNConv` operator receives a one-dimensional edge weight w_{ij}, it modulates the normalized adjacency. We can then compare the impact of distance as weights by computing 4 configurations: with and without edge weight and for each spatial view.

3.2 Graph Attention Network v2

Background. GATv2 [10] extends the original Graph Attention Network [11] by computing *dynamic* attention scores. It is basically an improved GCN where the message passing apply a weight determined by Attention on the neighbors. For each edge (i, j) of the layer ℓ, we first compute the non-normalised attention coefficient:

$$e_{ij}^{(\ell)} = \mathbf{a}^\top \mathrm{LeakyReLU}\big(\mathbf{W}[\,\mathbf{h}_i^{(\ell)} \| \mathbf{h}_j^{(\ell)}\,]\big), \tag{3}$$

where $\mathbf{a}$ et $\mathbf{W}$ are learnable weights. This score is then normalised to obtain the final attention score α_{ij} :

$$\alpha_{ij}^{(\ell)} = \frac{\exp(e_{ij}^{(\ell)})}{\sum_{k\in\mathcal{N}(i)} \exp(e_{ik}^{(\ell)})}. \tag{4}$$

Then, the node i of the next layer $\ell + 1$ is updated using this attention score :

$$\mathbf{h}_i^{(\ell+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(\ell)} \cdot \mathbf{W}\mathbf{h}_j^{(\ell)}\right). \tag{5}$$

Finally, to add the distance d_{ij} as a feature we can directly concatenate it with the non-normalized attention:

$$e_{ij}^{(\ell)} = \mathbf{a}^\top \mathrm{LeakyReLU}\big(\mathbf{W}[\,\mathbf{h}_i^{(\ell)} \| \mathbf{h}_j^{(\ell)} \| d_{ij}\,]\big). \tag{6}$$

Our GAT architecture is inspired by the *Epigraph* model [5], which demonstrated that GAT is effective for epitope prediction. Compared to the original Epigraph design we introduce four modifications: (i) GATv2 instead of simple GAT (ii) use of distance (iii) reduced number of layers to overcome possible over-smoothing (iv) application of a low dropout rate.

3.3 Multi-View Graph Attention Network

Background. A single graph topology captures only one aspect of residue proximity— either spatial contacts *or* sequential neighbors—yet both are biologically meaningful. The Multi-View Graph Attention Network (MGAT) [12] addresses this problem by learning separate embeddings on K distinct graph views and then merging them through an attention mechanism.

Each node i for the view k has its normalized attention coefficient β_{ik}, this coefficient is then used to weights each node representation $\mathbf{z}_{i,k}$:

The global node representation is:

$$\mathbf{h}_i^{\text{global}} = \sum_{k=1}^{K} \beta_{ik}\,\mathbf{z}_{i,k}\,. \tag{7}$$

A regularization term is also added to form the total loss in order to force the model to focus on different features:

$$\mathcal{L}_{total} = \mathcal{L}_{\text{task}} + \lambda\,\Omega \tag{8}$$

with $\lambda = 0.005$ and Ω the sum of the differences of the matrices between each residue i and j.

3.4 Equivariant Graph Neural Network

Background. Protein is invariant by translation and rotation, it means that applying these transformation on an antigen will not alter its conformational epitope. Standard GNNs lack this explicit inductive biais because they either ignore coordinates entirely or encode them as plain node features, breaking equivariance. The Equivariant Graph Neural Network (EGNN) [13] explicitly maintains *E(3)-equivariance*: node features are invariant to rotations and translations, while coordinate updates are equivariant.

Message passing. At each layer ℓ, the EGNN updates both the node embedding $\mathbf{h}_i$ and the coordinate $\mathbf{x}_i$. The concept is that the message passing $\mathbf{m}_{ij}$ depends on the square distance (an invariant scalar) and the euclidean distance:

$$\mathbf{m}_{ij} = \phi_e\big(\mathbf{h}_i^{(\ell)},\ \mathbf{h}_j^{(\ell)},\ \|\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)}\|^2,\ a_{ij}\big), \tag{9}$$

where ϕ_e is a two-layer MLP with SiLU activations and a_{ij} denotes the edge attribute distance, edge_dim $= 2$).

Coordinates update. Subsequently, the position of each residue is updated depending on the "force" of $\mathbf{m}_{ij}$. This mechanism allows the coordinates of each amino acid to evolve depending on the importance it gives to its neighbors. Consequently making the change of positions using the chemical properties hidden in the embeddings. The coordinates is weigthed by the message passing $\mathbf{m}_{ij}$:

$$\mathbf{x}_i^{(\ell+1)} = \mathbf{x}_i^{(\ell)} + \sum_{j\in\mathcal{N}(i)} \big(\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)}\big)\,\phi_x(\mathbf{m}_{ij}), \tag{10}$$

where ϕ_x is a two-layer MLP outputting a single scalar.

This approach shares conceptual similarities with GATv2, particularly because of how it computes neighbor coefficient depending on the features, however the equivariant property creates a dynamic chemical consistency that is not negligible. While GATv2 only sees the topological aspect of the graph, EGNN treats the protein as a 3D entity. It moves residues according to learned geometric rules and uses this displacement to reevaluate the importance of spatial proximity in subsequent layers, ensuring the joint evolution of message passing and coordinates.

Node update The aggregated messages are concatenated with the current node embedding and passed through a node-update MLP:

$$\mathbf{h}_i^{(\ell+1)} = \phi_h\Big(\mathbf{h}_i^{(\ell)},\ \sum_{j\in\mathcal{N}(i)} \mathbf{m}_{ij}\Big). \tag{11}$$

Attention-based variant. We also implemented a variant with **attention mechanism** into the EGNN message-passing framework. In the standard EGNN, all messages $\mathbf{m}_{ij}$ contribute equally to aggregation $\mathbf{h}_i^{(\ell+1)}$, we augment the architecture with a learned attention gate. This allows the model to selectively attend to the most informative neighbors while preserving equivariance, since α_{ij} is a scalar invariant (it depends only on distances and node features).

3.5 Architectures details

Table 1 provides a side-by-side comparison of the architectural characteristics of the four model families.

Table 1: Architectural comparison of the four GNN families.

	GCN	**GAT**	**MGAT**	**EGNN**
Layers	3	3	3×4 views	3
Hidden dim	128	128	128	128
Attention	No	Yes (4 heads)	Yes (4 heads)	Optional
Multi-view	No	No	Yes (4 views)	No
Equivariant	No	No	No	Yes
Coord. updates	No	No	No	Yes
Residual conn.	No	Yes	Yes (GATv2)	Yes
Dropout	0.3	0.1	0.1	0.1
Edge features	Optional (d_{ij})	Optional (d_{ij})	Optional (d_{ij})	d_{ij}

4 Results

We evaluate the models using standard classification metrics such as Area Under the Curve of Precision-Recall (AUC-PR), Area Under the Curve of Receiver Operating Characteristic (AUC-ROC), Matthews Correlation coefficient (MCC) and F1-score. The two most important metrics are MCC and AUC-PR because of their ability to represent performance under the use of an imbalanced dataset.

Table 2: Test performance. **Bold**: best, underline: second best.

Model	**AUC-PR**	**AUC-ROC**	**MCC**	**F1**
GCN				
Spatial$_1$	0.078 ± 0.007	0.681 ± 0.020	0.076 ± 0.012	0.082 ± 0.012
Spatial$_1$ + edge_w	0.080 ± 0.005	0.706 ± 0.009	0.073 ± 0.011	0.067 ± 0.012
Spatial$_2$	0.088 ± 0.006	0.722 ± 0.009	0.100 ± 0.009	0.095 ± 0.010
Spatial$_2$ + edge_w	0.087 ± 0.005	0.721 ± 0.011	0.086 ± 0.013	0.074 ± 0.016
GAT (GATv2)				
Spatial$_1$	0.249 ± 0.004	<u>0.773</u> ± 0.003	**0.238** ± 0.010	<u>0.289</u> ± 0.012
Spatial$_1$ + edge_w	0.243 ± 0.005	0.767 ± 0.003	<u>0.236</u> ± 0.012	**0.294** ± 0.014
Spatial$_2$	**0.258** ± 0.004	**0.773** ± 0.004	0.234 ± 0.012	0.285 ± 0.015
Spatial$_2$ + edge_w	0.234 ± 0.018	0.742 ± 0.027	0.194 ± 0.030	0.236 ± 0.029
EGNN (no attention)				
Spatial$_1$ / BCE	0.215 ± 0.008	0.692 ± 0.008	0.160 ± 0.013	0.245 ± 0.019
Spatial$_1$ / Focal	0.225 ± 0.005	0.701 ± 0.006	0.171 ± 0.013	0.247 ± 0.022
Spatial$_2$ / BCE	0.241 ± 0.003	0.763 ± 0.002	0.222 ± 0.010	0.278 ± 0.012
Spatial$_2$ / Focal	0.243 ± 0.005	0.763 ± 0.003	0.219 ± 0.012	0.279 ± 0.014
EGNN with attention				
Spatial$_1$ / BCE	<u>0.253</u> ± 0.004	0.713 ± 0.004	0.189 ± 0.009	0.281 ± 0.009
Spatial$_1$ / Focal	0.238 ± 0.007	0.699 ± 0.006	0.178 ± 0.007	0.266 ± 0.010
Spatial$_2$ / BCE	0.232 ± 0.005	0.753 ± 0.005	0.205 ± 0.009	0.262 ± 0.011
Spatial$_2$ / Focal	0.225 ± 0.005	0.750 ± 0.003	0.201 ± 0.008	0.261 ± 0.010
MGAT (4 views)				
MGAT$_1$ (edge + focal)	0.223 ± 0.006	0.748 ± 0.005	0.194 ± 0.016	0.246 ± 0.020
MGAT$_2$ (edge + BCE)	0.229 ± 0.006	0.744 ± 0.004	0.197 ± 0.010	0.243 ± 0.017
MGAT$_3$ (no edge + focal)	0.232 ± 0.005	0.751 ± 0.004	0.216 ± 0.009	0.276 ± 0.011
MGAT$_4$ (no edge + BCE)	0.226 ± 0.004	0.751 ± 0.003	0.204 ± 0.011	0.249 ± 0.023
External baselines (reported)				
GraphBepi	– ± –	0.620 ± –	0.160 ± –	0.280 ± –
EpiGraph	0.230 ± –	0.730 ± –	0.190 ± –	0.290 ± –

5 Discussion

5.1 Interpretation of the results

Effect of edge distance features GCN, GATv2, and MGAT that use this feature generally show equal or worse performance than models that do not use distance as weight on the edges. One of the main reasons might be that at a

molecular scale, the distance is not significant compared to the nature of electrostatic forces. A high distance could be easily compensated by a strong hydrogen bond, this implies that distance might confuse the models in their prediction.

Effect of the KNN approach It is difficult to draw a clear conclusion about the use of KNN graphs for all models. The GCN and the EGNN without attention take a clear advantage of the KNN approach, conversely the models using attention such as GATv2 and the enhanced EGNN do not show a clear trend. This is logical considering that Attention applies a neighbor selection, the non-attentive models suffer from the lack of this mechanism by weighting every neighbor as equal. However, the KNN graphs already perform a contextual neighbor ablation, helping the "classic" models, while slightly impacting negatively the attentive models.

Effect of the Focal Loss The use of Focal Loss as a loss function to replace Binary cross-entropy was supposed to address the imbalanced aspect of the dataset. Nevertheless, it did not increased the metrics and even showed worse performance in some cases. The use of fixed hyper-parameters might be one of the reason, tuning α and γ should reveal a stronger trend in its impact on the models.

Effect of the model GATv2 models achieved the highest performance across all of the metrics primarily due to their dynamic neighbor selection mechanism. Interestingly, we observe that MGATs achieved worse performance while combining several GATs. The main reason is that only one spatial GAT can already understand the local environment, then the other graphs will confuse the prediction. Our "classic" EGNN showed competitive results by using its chemical consistency based mechanism, and the add of dynamic Attention has increased the performances of two spatial_1 models, while making worse the two spatial_2 graph based models. This confirms the intuition that weighting neighbors on its features using Attention disturbs the prediction when using a KNN graph-based, but helps situation with a lot of neighbors, such as depicted by spatial_1 graphs.

5.2 Limitations and Future Work

State-of-the-art models are now focusing on a more precise problem: Ab specific epitope prediction [14]. While we are adopting an Ab-agnostic approach, the Ab specific paradigm has several benefits and its applications are more direct. Instead of looking at an Ag in order to predict the positions of epitope, it searches for an Ag that can bind to a specific Ab. It allows to directly analyze the specificity of an Ab while our method is more general and shows trends in the epitope position. However our work remains relevant to study the particularities of GNNs through this difficult problem. Understanding the trends of epitopes using the most used Deep Learning framework is still interesting as it provides a benchmark for how GNNs encode complex geometric and chemical patterns from PLMs embeddings.

The dataset we used has a reduced number of protein for training (180), and this limits our models generalization. We kept the loss function very simple for ablation purpose, however adding regularization techniques exploiting the homophily property of epitope or addressing the imbalanced aspect of the dataset might drastically increase the performances. The use of Radial Basis Function to enhance the consideration of distance in the embeddings could also give stronger trends than only using a raw scalar.

6 Conclusion

Neighbor selectivity appears to be key in the prediction performance of epitope regions. GATv2 with its dynamic attention showed greater performances than the GCN, showing not only how the local environment matters, but also that at this scale, precision in neighbors feature attention is crucial. Similarly, the use of a graph based on the K-nearest neighbors seems to enhance the prediction for the non-attentive models, while showing less indicative trends for the models using Attention mechanisms. Euclidean distance is a misleading edge feature, incorporating raw Euclidean distance tends to worsen or fail to improve overall performance across GCN, GATv2 and MGAT models. We hypothesized that this is a consequence of the negligible role of distance at this molecular scale. Finally it is hard to draw a strong conclusion as the results are inconsistent, tuning its hyper-parameters should show a more reliable trend.

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